NORTH WESTERN RAILWAY

Serving customers with a smile — GDCE Practice Question Paper

Test Date	27/08/2026	**Test Time**	1:00 PM – 3:00 PM
Test Center	Dr Tanveer Malawat Nursing College Bikaner		
Subject	JE Civil and Allied Engineering		

SECTION: JE COMMON

Q.1 Crop rotation and use of organic fertilizers help in reducing soil pollution on agricultural land. This is because these practices:

1. Revitalize soil and reduce chemical accumulation
2. Increase chemical accumulation
3. Promote soil salinity
4. Decrease organic matter

`Question ID: 4410091955054` `Opt 1: 4410097700992` `Opt 2: 4410097700990` `Opt 3: 4410097700991` `Opt 4: 4410097700993`

Q.2 Which of the following statements is INCORRECT for mass number?

1. Protium and deuterium have same the mass number.
2. Isobars have same mass number.
3. The number of nucleons is same as mass number.
4. Mass number is equal to the sum of number of neutrons and number of protons.

`Question ID: 441009909322` `Opt 1: 4410093591148` `Opt 2: 4410093591147` `Opt 3: 4410093591146` `Opt 4: 4410093591145`

Q.3 Which of the following is an example of system software?

1. Operating System
2. Google Chrome
3. MS Excel
4. Adobe Photoshop

`Question ID: 4410091555490` `Opt 1: 4410096144519` `Opt 2: 4410096144518` `Opt 3: 4410096144517` `Opt 4: 4410096144520`

Q.4 What is the primary role of network security in a computing environment?

1. Detecting software errors and fixing bugs
2. Making the computer faster by optimizing resources
3. Logging user activity for reporting purposes
4. Blocking unauthorized access in networks and securing data transmission

Question ID: 4410091048034 Opt 1: 4410094141113 Opt 2: 4410094141116 Opt 3: 4410094141114 Opt 4: 4410094141115

Q.5 An industrial unit is evaluating strategies to limit greenhouse gas emissions. Which approach would be best to effectively reduce emissions and mitigate climate change impact?

1. Switching to renewable energy sources and improving efficiency
2. Regular maintenance of industrial equipment
3. Increasing the greenbelt area around the unit
4. Installing air filtration systems at the exhaust

Question ID: 4410091946019 Opt 1: 4410097665542 Opt 2: 4410097665540 Opt 3: 4410097665541 Opt 4: 4410097665539

Q.6 A 2 kg body accelerates at 3 m/s². The force acting on it is:

1. 2 N
2. 6 N
3. 3 N
4. 5 N

Question ID: 4410091369526 Opt 1: 4410095409560 Opt 2: 4410095409558 Opt 3: 4410095409561 Opt 4: 4410095409559

Q.7 Which of the following metals is the least reactive?

1. Magnesium
2. Gold
3. Zinc
4. Sodium

Question ID: 441009908760 Opt 1: 4410093588895 Opt 2: 4410093588897 Opt 3: 4410093588896 Opt 4: 4410093588898

Q.8 Which of the following schemes was announced by the Government of India in Union Budget 2026-27?

1. Agnipath Scheme
2. PM Vishwakarma Scheme
3. India Semiconductor Mission 2.0
4. PM SHRI Schools

Question ID: 4410092283427 Opt 1: 4410098982831 Opt 2: 4410098982834 Opt 3: 4410098982833 Opt 4: 4410098982832

Q.9 Which of the following statements is INCORRECT regarding average velocity?

1. It is always equal to average speed.
2. It is measured in m/s.
3. It is obtained by dividing total displacement by total time.
4. It depends on the direction of motion.

Question ID: 4410091375421 Opt 1: 4410095432996 Opt 2: 4410095432995 Opt 3: 4410095432994 Opt 4: 4410095432997

Q.10 Which popular slogan was given by Bal Gangadhar Tilak to arouse nationalist feelings among Indians?

1. Swaraj is my birthright and I shall have it
2. Inquilab Zindabad
3. Quit India
4. Vande Mataram

Question ID: 441009876064 Opt 1: 4410093457870 Opt 2: 4410093457869 Opt 3: 4410093457867 Opt 4: 4410093457868

Q.11 Which of the following has straight-chain hydrocarbons?

1. 2-Methylpropane
2. Butane
3. Isobutane
4. 3-Methylpentane

Question ID: 4410091026137 Opt 1: 4410094056120 Opt 2: 4410094056119 Opt 3: 4410094056121 Opt 4: 4410094056122

Q.12 What is the primary function of the Public Accounts Committee (PAC) of the Indian Parliament?

1. Evaluating government schemes
2. Assisting in the election of the Speaker
3. Recommending the central budget
4. Reviewing the audit reports of CAG

Question ID: 441009876003 Opt 1: 4410093457630 Opt 2: 4410093457632 Opt 3: 4410093457629 Opt 4: 4410093457631

Q.13 Lakhwar Multipurpose Project is related to which state?

1. Uttar Pradesh
2. Uttarakhand
3. Rajasthan
4. Madhya Pradesh

Question ID: 441009882640 Opt 1: 4410093484241 Opt 2: 4410093484239 Opt 3: 4410093484240 Opt 4: 4410093484238

Q.14 In July 2026, in which city did Bihar Chief Minister Samrat Choudhary launch the Bihar Heli-Tourism and Air Tourism Service Scheme?

1. Gaya
2. Rajgir
3. Patna
4. Bhagalpur

Question ID: 4410092283461 Opt 1: 4410098982967 Opt 2: 4410098982970 Opt 3: 4410098982969 Opt 4: 4410098982968

Q.15 Vijayanagara architecture is known for incorporating Hindu and Islamic elements. Which option best represents this architectural synthesis?

1. Use of granite pillars and carved stone halls only
2. Avoidance of all foreign architectural and decorative elements
3. Construction of temple complexes only without secular buildings
4. Traditional Dravidian structures with domes and arches

Question ID: 4410092283611 Opt 1: 4410098983564 Opt 2: 4410098983565 Opt 3: 4410098983566 Opt 4: 4410098983563

Q.16 In MS PowerPoint, which tab provides tools to insert visual elements such as diagrams, illustrations, and graphical content into a slide?

1. Review
2. Insert
3. Home
4. Design

Question ID: 441009595550 Opt 1: 4410092337320 Opt 2: 4410092337319 Opt 3: 4410092337317 Opt 4: 4410092337318

Q.17 In MS Excel, how do you create a text series (e.g., "A", "B", "C")?

1. By using the Fill command from the Home tab
2. By using the Insert tab to create a text series
3. By manually typing text into each cell
4. By typing "A" in the first cell, then holding Ctrl and dragging the fill handle

Question ID: 441009936208 Opt 1: 4410093698588 Opt 2: 4410093698590 Opt 3: 4410093698589 Opt 4: 4410093698587

Q.18 During which season was the National Agricultural Insurance Scheme launched to protect farmers from crop losses due to natural calamities like droughts, floods, hailstorms, cyclones, pests, and diseases?

1. Rabi 2005-2006
2. Kharif 1998-1999
3. Kharif 2001-2002
4. Rabi 1999-2000

Question ID: 441009878678 Opt 1: 4410093468368 Opt 2: 4410093468365 Opt 3: 4410093468367 Opt 4: 4410093468366

Q.19 When iron filings are mixed with sulfur powder and heated, a black substance called iron sulfide is formed. Which statement gives the most accurate description of these changes?

1. Iron dissolves in sulfur to form a solution.
2. Merely mixing the substances causes a physical change.
3. A reversible physical change takes place.
4. A chemical change occurs resulting in the formation of a new substance.

Question ID: 4410091606439 Opt 1: 4410096344770 Opt 2: 4410096344768 Opt 3: 4410096344769 Opt 4: 4410096344771

Q.20 Mudumalai National Park in Tamil Nadu, sharing boundaries with Kerala and Karnataka, is part of which Biosphere Reserve in India?

1. Agasthyamala Biosphere Reserve
2. Nilgiri Biosphere Reserve
3. Nokrek Biosphere Reserve
4. Khangchendzonga Biosphere Reserve

Question ID: 441009883795 Opt 1: 4410093489149 Opt 2: 4410093489146 Opt 3: 4410093489147 Opt 4: 4410093489148

Q.21 Which of the following default TCP port pairs is correct for standard/secure communication?

1. FTP-8080, HTTPS-110
2. IMAP-22, HTTP-25
3. HTTP-80, HTTPS-443
4. SMTP-21, POP3-23

Question ID: 4410091434879 Opt 1: 4410095668068 Opt 2: 4410095668067 Opt 3: 4410095668065 Opt 4: 4410095668066

Q.22 The 'Bharat-VISTAAR' scheme announced in Union Budget 2026-27 belongs to which Central Ministry?

1. Ministry of Environment, Forest and Climate Change
2. Ministry of Education
3. Ministry of Housing and Urban Affairs
4. Ministry of Agriculture and Farmers Welfare

Question ID: 4410092283456 Opt 1: 4410098982948 Opt 2: 4410098982947 Opt 3: 4410098982950 Opt 4: 4410098982949

Q.23 The Vienna Convention for the Protection of the Ozone Layer (1985) paved the way for which upcoming international treaty?

1. Paris Agreement
2. Basel Convention
3. Montreal Protocol
4. Kyoto Protocol

Question ID: 4410092283623 Opt 1: 4410098983611 Opt 2: 4410098983614 Opt 3: 4410098983613 Opt 4: 4410098983612

Q.24 In which of the following years did Vasco da Gama land at Calicut on the Malabar Coast for the second time?

1. 1511
2. 1509
3. 1505
4. 1502

Question ID: 441009846067 Opt 1: 4410093337148 Opt 2: 4410093337147 Opt 3: 4410093337146 Opt 4: 4410093337145

Q.25 What is the chemical formula of Plaster of Paris?

1. $CaSO_4$
2. $CaCO_3$
3. $CaSO_4 \cdot 2H_2O$
4. $CaSO_4 \cdot \frac{1}{2}H_2O$

Question ID: 441009940195 Opt 1: 4410093714430 Opt 2: 4410093714431 Opt 3: 4410093714428 Opt 4: 4410093714429

Q.26 A new waste-to-energy plant is proposed near a residential area. What would be the primary environmental concern related to this facility?

1. Traffic congestion due to waste transportation
2. Leachate generation affecting groundwater
3. Excessive noise pollution generated by machinery
4. Emission of toxic gases during incineration

Question ID: 4410091953112 Opt 1: 4410097693342 Opt 2: 4410097693344 Opt 3: 4410097693345 Opt 4: 4410097693343

Q.27 A body moves in a straight line with variable speed. Its displacement in 10 s is 50 m. What will be its average velocity?

1. 25 m/s
2. 50 m/s
3. 5 m/s
4. 10 m/s

Question ID: 4410091317505 Opt 1: 4410095201582 Opt 2: 4410095201581 Opt 3: 4410095201579 Opt 4: 4410095201580

Q.28 A textile factory discharges untreated dye-laden water into a nearby stream. What is the most direct environmental consequence of this action?

1. Rapid growth of large aquatic plants
2. Reduction in aquatic life diversity
3. Increase in water temperature
4. Increase in water salinity

Question ID: 4410091957918 Opt 1: 4410097712309 Opt 2: 4410097712306 Opt 3: 4410097712307 Opt 4: 4410097712308

Q.29 Which of the following is a socio-economic Directive Principle of State Policy?

1. Equal pay for equal work
2. Separation of powers
3. Independence of judiciary
4. Promotion of international peace

Question ID: 441009902746 Opt 1: 4410093564796 Opt 2: 4410093564795 Opt 3: 4410093564797 Opt 4: 4410093564794

Q.30 Frequent algal blooms occur in a nearby lake. Which activity in the surrounding area is responsible for this?

1. Discharging untreated sewage into the lake
2. Using sandbags for flood control
3. Constructing stone pathways around the lake
4. Planting native trees along the lake bank

Question ID: 4410091957963 Opt 1: 4410097712489 Opt 2: 4410097712487 Opt 3: 4410097712488 Opt 4: 4410097712486

Q.31 Which statement regarding Cache Memory is INCORRECT?

1. Cache memory stores frequently accessed data for fast retrieval.
2. Cache memory is located inside or close to the CPU.
3. Cache memory is a form of secondary storage.
4. Cache memory is faster than main memory.

Question ID: 441009186258 Opt 1: 441009736248 Opt 2: 441009736247 Opt 3: 441009736249 Opt 4: 441009736246

Q.32 What is the use of the Insert (Ins) key on a keyboard?

1. Pastes content
2. Toggles between Insert and Overwrite modes
3. Deletes inserted text
4. Refreshes the screen

Question ID: 441009141188 Opt 1: 441009560968 Opt 2: 441009560970 Opt 3: 441009560969 Opt 4: 441009560971

Q.33 Which statement is correct regarding the Right to Property in India?

1. It is a fundamental right under Article 31 (now repealed).
2. It was completely abolished by the 42nd Amendment (1976).
3. It is no longer a Fundamental Right.
4. It is a Fundamental Right under Article 32 of the Constitution.

`Question ID: 441009902760` `Opt 1: 4410093564851` `Opt 2: 4410093564853` `Opt 3: 4410093564852` `Opt 4: 4410093564850`

Q.34 A plan is being made to reduce urban air pollution by planting a greenbelt along the edge of a city's main industrial area. What mechanism is used in the greenbelt to improve air quality?

1. Dilution of pollutants through increased wind speed
2. Increased rainfall to wash off pollutants
3. Reduction of industrial emissions at source
4. Absorption of gaseous pollutants by plant leaves

`Question ID: 4410091945981` `Opt 1: 4410097665389` `Opt 2: 4410097665387` `Opt 3: 4410097665390` `Opt 4: 4410097665388`

Q.35 In MS Word, what should you do to remove bullets from a list?

1. Press Backspace
2. Click the Bullets button again
3. Highlight text and cut
4. Manually delete each bullet

`Question ID: 441009249138` `Opt 1: 441009970467` `Opt 2: 441009970466` `Opt 3: 441009970468` `Opt 4: 441009970465`

Q.36 Which operating system concept is most directly involved in ensuring that multiple processes can access and modify files simultaneously without corrupting data?

1. Virtual memory management
2. Scheduling algorithms
3. Paging
4. File locking mechanisms

`Question ID: 441009280727` `Opt 1: 4410091090830` `Opt 2: 4410091090828` `Opt 3: 4410091090827` `Opt 4: 4410091090829`

Q.37 Stars appear to twinkle more clearly when:

1. They are at their zenith
2. The sky is overcast with clouds
3. The moon is bright
4. They are near the horizon

`Question ID: 4410091350003` `Opt 1: 4410095332156` `Opt 2: 4410095332155` `Opt 3: 4410095332158` `Opt 4: 4410095332157`

Q.38 The Non-Cooperation Movement of 1920 included all of the following EXCEPT:

1. Violent attacks on officials
2. Boycott of schools and colleges
3. Refusal to pay taxes
4. Boycott of British courts

`Question ID: 441009876180` `Opt 1: 4410093458333` `Opt 2: 4410093458334` `Opt 3: 4410093458332` `Opt 4: 4410093458331`

Q.39 Which of the following analogies was used by J.J. Thomson to describe his atomic model?

1. Christmas pudding
2. Planetary system
3. Nuclear core-shell
4. Solar system

`Question ID: 441009940962` `Opt 1: 4410093717512` `Opt 2: 4410093717510` `Opt 3: 4410093717509` `Opt 4: 4410093717511`

Q.40 Match the following computer security terms with their correct definitions:

Term	Definition
1. Encryption	A. Blocks unauthorized network access
2. Firewall	B. Validates user identity
3. Antivirus	C. Converts data into unreadable code
4. Authentication	D. Detects and removes malware

1. 1-C, 2-A, 3-D, 4-B
2. 1-C, 2-A, 3-B, 4-D
3. 1-D, 2-C, 3-A, 4-B
4. 1-A, 2-B, 3-D, 4-C

`Question ID: 4410091820298` `Opt 1: 4410097176426` `Opt 2: 4410097176427` `Opt 3: 4410097176425` `Opt 4: 4410097176424`

Q.41 Which statement is TRUE regarding parallel connection of resistors?

1. Total current is the sum of currents flowing through individual resistors.
2. Total current is equal to the current through any single branch.
3. Adding more resistors increases the total equivalent resistance.
4. Each resistor carries equal current regardless of resistance value.

`Question ID: 4410091367660` `Opt 1: 4410095402079` `Opt 2: 4410095402077` `Opt 3: 4410095402078` `Opt 4: 4410095402080`

Q.42 According to Bohr's atomic model, electrons revolve around the nucleus in:

1. Straight lines
2. Discrete orbits
3. Elliptical paths
4. Random paths

`Question ID: 441009982706` `Opt 1: 4410093883283` `Opt 2: 4410093883280` `Opt 3: 4410093883281` `Opt 4: 4410093883282`

Q.43 Which of the following is a heterogeneous mixture?

1. Salt solution
2. Air
3. Sand and water
4. Sugar solution

`Question ID: 4410091624576` `Opt 1: 4410096416835` `Opt 2: 4410096416837` `Opt 3: 4410096416838` `Opt 4: 4410096416836`

Q.44 Which of the following statements are correct?

Statement I: In an open pit design transfer station, collection vehicles dump waste into an open pit, from where waste material is mechanically or manually transferred into larger open hoppers.
Statement II: In a direct transfer station, waste from smaller vehicles is transferred directly into larger vehicles.

1. Only Statement II is correct
2. Both Statement I and II are correct
3. Neither Statement I nor Statement II is correct
4. Only Statement I is correct

Question ID: 4410091989697 Opt 1: 4410097836546 Opt 2: 4410097836547 Opt 3: 4410097836548 Opt 4: 4410097836545

Q.45 Who wrote the book 'The Second Orbit' published in June 2026?

1. Anil Joshi
2. Priya Sharma
3. Rohit Kumar
4. Shubhanshu Shukla

Question ID: 4410092292561 Opt 1: 4410099016715 Opt 2: 4410099016717 Opt 3: 4410099016714 Opt 4: 4410099016716

Q.46 Fill in the blank: The generation of foul-smelling gases in unmanaged landfills is primarily due to _____.

1. Oxidation reactions
2. Anaerobic decomposition
3. Aerobic decomposition
4. Photosynthesis

Question ID: 4410091953117 Opt 1: 4410097693365 Opt 2: 4410097693363 Opt 3: 4410097693362 Opt 4: 4410097693364

Q.47 _____ is an air pollutant that binds with hemoglobin in blood and impairs oxygen supply.

1. Nitrogen dioxide
2. Sulfur dioxide
3. Carbon monoxide
4. Ozone

Question ID: 4410091988061 Opt 1: 4410097830024 Opt 2: 4410097830023 Opt 3: 4410097830022 Opt 4: 4410097830021

Q.48 A real and inverted image is formed at a distance of 30 cm from a concave mirror when the object is placed 40 cm from the mirror. What will be the magnification produced by the mirror?

1. +3/4
2. -4/3
3. +4/3
4. -3/4

Question ID: 4410091376217 Opt 1: 4410095436430 Opt 2: 4410095436429 Opt 3: 4410095436428 Opt 4: 4410095436431

Q.49 Why are ultrasounds preferred for cleaning objects with complex shapes?

1. They have very high speed in liquids
2. Their high frequency dislodges dirt particles
3. They increase temperature of solution
4. They produce magnetic effect

Question ID: 4410091367716 Opt 1: 4410095402298 Opt 2: 4410095402299 Opt 3: 4410095402297 Opt 4: 4410095402300

Q.50 Which method is commonly used for the safe disposal of infectious biomedical waste?

1. Composting
2. Landfilling
3. Recycling
4. Incineration

Question ID: 4410091953115 Opt 1: 4410097693350 Opt 2: 4410097693352 Opt 3: 4410097693351 Opt 4: 4410097693353

SECTION: JE CIVIL AND ALLIED ENGINEERING

Q.1 Which surveying activity is best performed using a Total Station?

1. Estimating concrete mix maps
2. Plotting topographic maps
3. Testing asphalt properties
4. Borehole logging maps

Question ID: 4410091078241 Opt 1: 4410094250319 Opt 2: 4410094250320 Opt 3: 4410094250321 Opt 4: 4410094250322

Q.2 Which of the following plastics is used in fiber-reinforced composites for bridges and structural applications?

1. Acrylic
2. Acrylonitrile Butadiene Styrene
3. Epoxy resin
4. Nylon

Question ID: 44100930943 Opt 1: 441009123260 Opt 2: 441009123263 Opt 3: 441009123262 Opt 4: 441009123261

Q.3 Cadastral surveying is mainly conducted for:

1. Designing dams and canals
2. Surveying mountains
3. Preparing contour maps
4. Fixing property boundaries and land ownership

Question ID: 4410091363034 Opt 1: 4410095384073 Opt 2: 4410095384072 Opt 3: 4410095384071 Opt 4: 4410095384074

Q.4 According to the Mohr-Coulomb failure criterion, which of the following conditions must be met for a soil element to be considered at the point of shear failure?

1. The shear stress τ on the plane of maximum obliquity equals the normal stress σ on plane.
2. The Mohr circle representing the state of stress becomes tangent to the failure envelope.
3. The major principal stress σ_1 reaches its peak value.
4. The effective stress σ' on the failure plane reaches zero, leading to a quick condition.

Question ID: 4410091405505 Opt 1: 4410095551161 Opt 2: 4410095551162 Opt 3: 4410095551160 Opt 4: 4410095551163

Q.5 What is the primary cause of rail creep?

1. Wave action of rail under dynamic moving loads
2. Excessive ballast cushion
3. Temperature variation
4. Defective fastenings only

Question ID: 4410091243207 Opt 1: 4410094906165 Opt 2: 4410094906167 Opt 3: 4410094906164 Opt 4: 4410094906166

Q.6 Which of the following is the best method to correct for local attraction in a traverse?

1. Use magnetic declination tables
2. Use a prismatic compass only
3. Avoid metal objects during survey
4. Apply correction based on included angles

Question ID: 4410091078410 Opt 1: 4410094250978 Opt 2: 4410094250975 Opt 3: 4410094250977 Opt 4: 4410094250976

Q.7 Which type of uniform thickness doors are constructed using plywood or block board?

1. Collapsible doors
2. Framed and panelled doors
3. Rolling shutters
4. Flush doors

Question ID: 44100970789 Opt 1: 441009281764 Opt 2: 441009281765 Opt 3: 441009281763 Opt 4: 441009281766

Q.8 The shape of a bending moment diagram under UDL is:

1. Linear
2. Triangular
3. Parabolic
4. Constant

`Question ID: 441009224099` `Opt 1: 441009877682` `Opt 2: 441009877685` `Opt 3: 441009877683` `Opt 4: 441009877684`

Q.9 For a short column, which of the following is true regarding the value of P_E (crippling load obtained by Euler's formula) compared to the value of P_R (crippling load obtained by Rankine's formula)?

1. P_E is very high, and the Rankine formula gives a value close to P_E
2. P_E is very small, and P_R is close to the ultimate crushing load.
3. P_E is very high, and P_R is close to the ultimate crushing load.
4. P_E is quite small, and Rankine's formula provides a very different value compared to P_E.

`Question ID: 44100926948` `Opt 1: 441009107369` `Opt 2: 441009107370` `Opt 3: 441009107371` `Opt 4: 441009107372`

Q.10 Which factor ensures future expansion of the bridge while selecting the site?

1. Selecting suitable type of bearings for superstructure
2. Choosing appropriate colour of parapets and railings
3. Providing sufficient clearance for navigation and traffic
4. Deciding adequate number of handrails for pedestrians

`Question ID: 4410091604090` `Opt 1: 4410096335547` `Opt 2: 4410096335548` `Opt 3: 4410096335546` `Opt 4: 4410096335549`

Q.11 The working principle of a vibratory roller is based on:

1. Generation of air suction inside the drum during rolling action
2. Hydraulic pressure applied directly on the drum surface
3. Conversion of static load into centrifugal force by rotating weights
4. Pneumatic compression produced by tyres mounted on axle

Question ID: 4410091291601 Opt 1: 4410095098431 Opt 2: 4410095098432 Opt 3: 4410095098430 Opt 4: 4410095098433

Q.12 Which type of elevator is best suited for high-rise buildings?

1. Hydraulic elevator
2. Pneumatic elevator
3. Belt elevator
4. Traction elevator

Question ID: 44100921919 Opt 1: 44100987399 Opt 2: 44100987401 Opt 3: 44100987402 Opt 4: 44100987400

Q.13 What is the primary purpose of the flexural equation in structural analysis?

1. To calculate the maximum bending stress and design beams for bending loads
2. To evaluate the axial load capacity of columns under bending forces
3. To calculate the deflection at various points on the beam under load
4. To determine the shear force and bending moment at a specific section of the beam

Question ID: 44100926879 Opt 1: 441009107101 Opt 2: 441009107103 Opt 3: 441009107104 Opt 4: 441009107102

Q.14 Which rock exploration method is most suitable for obtaining continuous undisturbed soil samples from highly resistant strata for detailed geotechnical analysis?

1. Rotary drilling with core barrel
2. Auger Boring (Continuous Flight Auger)
3. Wash Boring
4. Percussion Drilling with a chisel bit

Question ID: 4410091476461 Opt 1: 4410095833151 Opt 2: 4410095833150 Opt 3: 4410095833152 Opt 4: 4410095833153

Q.15 Which of the following is NOT a type of combined footing in shallow foundations?

1. Rectangular
2. Strap
3. Trapezoidal
4. Mat

Question ID: 4410091076705 Opt 1: 4410094244237 Opt 2: 4410094244236 Opt 3: 4410094244235 Opt 4: 4410094244234

Q.16 Which of the following is NOT an intersection type on highways?

1. At-grade
2. Grade-separated
3. Rotary
4. Highway bypass without crossings

Question ID: 4410091470870 Opt 1: 4410095811423 Opt 2: 4410095811424 Opt 3: 4410095811425 Opt 4: 4410095811426

Q.17 In which brick bond are the bricks NOT placed at an angle?

1. Herringbone bond
2. Zigzag bond
3. English cross bond
4. Diagonal bond

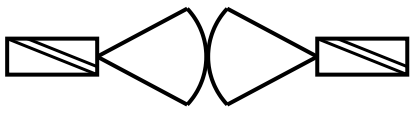
Question ID: 441009308619 Opt 1: 4410091201159 Opt 2: 4410091201158 Opt 3: 4410091201156 Opt 4: 4410091201157

Q.18 A tension member with bolted connections is most likely to fail:

1. by net cross-section rupture
2. at lateral supports
3. by gross cross-section twisting
4. by mid-span due to bending

Question ID: 4410091339335 Opt 1: 4410095289415 Opt 2: 4410095289418 Opt 3: 4410095289416 Opt 4: 4410095289417

Q.19 Identify the type of door symbol shown below, used in a building drawing as per IS 962 (1989):



1. Double-leaf sliding door unit
2. Double-leaf double-swing door
3. Single-leaf single-swing door
4. Single-leaf pivoted door unit

Question ID: 4410091528967 Opt 1: 4410096039001 Opt 2: 4410096038999 Opt 3: 4410096039000 Opt 4: 4410096039002

Q.20 Which of the following is a key step in the manufacturing process of fly ash bricks?

1. Moulding with clay slurry
2. Firing in a blast furnace
3. Curing with acid solution
4. Mixing fly ash, lime and gypsum

Question ID: 4410091320826 Opt 1: 4410095214865 Opt 2: 4410095214862 Opt 3: 4410095214864 Opt 4: 4410095214863

Q.21 Units of measurement for a drawing in AutoCAD can be set up using the command:

1. Limits
2. Grid
3. Scale
4. Units

Question ID: 44100936338 Opt 1: 441009144866 Opt 2: 441009144867 Opt 3: 441009144868 Opt 4: 441009144865

Q.22 Calculate the approximate cost of a building having a plinth area of 500 m² and a height of 4 m, assuming the rate per cubic content is ₹3,000.

1. ₹1,20,000
2. ₹6,00,000
3. ₹12,00,000
4. ₹60,00,000

Question ID: 44100929236 Opt 1: 441009116425 Opt 2: 441009116424 Opt 3: 441009116423 Opt 4: 441009116422

Q.23 What is the primary purpose of constructing pilot tunnels or exploratory headings?

1. To install permanent lining for the full section of the tunnel
2. To reduce operational speed of TBM during excavation
3. To enlarge tunnel diameter for larger trains or traffic
4. To confirm sub-surface conditions before commencing full tunnel excavation

Question ID: 4410091604466 Opt 1: 4410096337037 Opt 2: 4410096337036 Opt 3: 4410096337035 Opt 4: 4410096337034

Q.24 In an RCC cantilever beam subjected to various point loads and UDL directed downwards, the flexural tension reinforcement is provided:

1. at the center of the cross-section
2. at both the top and the bottom beam section
3. above the neutral axis of the beam section
4. below the neutral axis of the beam section

Question ID: 441009311375 Opt 1: 4410091211920 Opt 2: 4410091211921 Opt 3: 4410091211918 Opt 4: 4410091211919

Q.25 Which of the following is a function of the strap in Strap Footing?

1. To connect the wall with column footing
2. To transfer column loads to soil below both footings equally
3. To support the load of the wall above it
4. To transfer the structural load of columns to the soil directly

Question ID: 44100945151 Opt 1: 441009179815 Opt 2: 441009179816 Opt 3: 441009179813 Opt 4: 441009179814

Q.26 In a float-type rain gauge, the rainfall is measured by:

1. Weight of water
2. The number of bucket tips
3. Manual measurement
4. Rise in water level inside a float chamber

Question ID: 441009933588 Opt 1: 4410093688108 Opt 2: 4410093688106 Opt 3: 4410093688109 Opt 4: 4410093688107

Q.27 Which statement regarding fineness modulus is correct as per Indian Standards?

1. Fineness modulus is used for grading zone classification in IS 383
2. Fineness modulus replaces grading requirements in IS 383
3. IS 383 mandates fineness modulus limits for fine aggregates
4. Fineness modulus is an indicative index derived from sieve analysis

Question ID: 4410091401420 Opt 1: 4410095534993 Opt 2: 4410095534994 Opt 3: 4410095534991 Opt 4: 4410095534992

Q.28 A circular steel column has a diameter of 200 mm and an effective length of 4 m. Find its slenderness ratio.

1. 40
2. 80
3. 100
4. 60

Question ID: 4410091301577 Opt 1: 4410095138518 Opt 2: 4410095138520 Opt 3: 4410095138521 Opt 4: 4410095138519

Q.29 For a roadside obstruction located on a curve, the object marking should be:

1. Illuminated with a flashing lamp
2. Vertical and retro-reflective
3. Painted in dull colors without reflectivity
4. Horizontal with warning stripes

Question ID: 4410091350284 Opt 1: 4410095333272 Opt 2: 4410095333270 Opt 3: 4410095333273 Opt 4: 4410095333271

Q.30 Which of the following is the main moving component in a reciprocating pump?

1. Suction pipe
2. Delivery pipe
3. Piston
4. Cylinder

Question ID: 441009257666 Opt 1: 4410091004502 Opt 2: 4410091004503 Opt 3: 4410091004501 Opt 4: 4410091004500

Q.31 For a cantilever beam with UDL, the shear force diagram is:

1. Linearly varying and zero at free end
2. Parabolic and zero at free end
3. Parabolic and zero at fixed end
4. Constant from free end to fixed end

Question ID: 4410091527557 Opt 1: 4410096033404 Opt 2: 4410096033406 Opt 3: 4410096033405 Opt 4: 4410096033403

Q.32 Which of the following is the first and most critical step in the painting process?

1. Cleaning the brushes
2. Applying the topcoat
3. Choosing the paint colour
4. Surface preparation

Question ID: 4410091173154 Opt 1: 4410094626776 Opt 2: 4410094626773 Opt 3: 4410094626775 Opt 4: 4410094626774

Q.33 A simply supported Warren truss of 10 m span carries a UDL of 10 kN/m on its top chord. The reactions at supports (assume pin and roller supports) at left and right are respectively:

1. 40 kN and 10 kN
2. 50 kN and 50 kN
3. 25 kN and 25 kN
4. 30 kN and 20 kN

Question ID: 4410091272741 Opt 1: 4410095022978 Opt 2: 4410095022979 Opt 3: 4410095022976 Opt 4: 4410095022977

Q.34 Which dewatering/drainage method is most suitable for fine-grained cohesive soils?

1. Deep well system
2. Electro-Osmosis method
3. Shallow well system
4. Well point system

Question ID: 44100945175 Opt 1: 441009179902 Opt 2: 441009179903 Opt 3: 441009179904 Opt 4: 441009179901

Q.35 Which principle involves converting waste materials into new products?

1. Reuse
2. Recycle
3. Reduce
4. Recover

Question ID: 4410091045223 Opt 1: 4410094129971 Opt 2: 4410094129972 Opt 3: 4410094129970 Opt 4: 4410094129973

Q.36 A launching girder is a special-purpose _____ used in bridge construction, specifically in segmental bridges utilizing precast box girder bridge segments or precast girders in highway and high-speed rail projects.

1. Tower crane
2. Overhead crane
3. Crawler crane
4. Mobile gantry crane

Question ID: 4410091201398 Opt 1: 4410094739377 Opt 2: 4410094739379 Opt 3: 4410094739378 Opt 4: 4410094739376

Q.37 In a California Bearing Ratio (CBR) test, the recorded penetration resistance of a soil sample is 325 kg at 2.5 mm penetration and 723.5 kg at 5 mm penetration. Based on standard CBR test procedures, what is the design CBR value for this soil? (Round off to nearest whole number)

1. 15%
2. 35%
3. 25%
4. 45%

Question ID: 441009970239 Opt 1: 4410093834102 Opt 2: 4410093834101 Opt 3: 4410093834100 Opt 4: 4410093834103

Q.38 The arrow head in a force representation indicates:

1. the magnitude of the force
2. the starting point of the force
3. the direction in which the force acts
4. the type of force applied

Question ID: 44100962112 Opt 1: 441009247475 Opt 2: 441009247474 Opt 3: 441009247476 Opt 4: 441009247477

Q.39 In planning a residence, the principle of orientation is adopted to ensure:

1. beautiful aesthetic appearance of the residence
2. rooms are oriented to achieve the required privacy
3. natural elements are used effectively in the design
4. proper sanitation and hygiene in the residence

Question ID: 44100936113 Opt 1: 441009143983 Opt 2: 441009143981 Opt 3: 441009143980 Opt 4: 441009143982

Q.40 Identify the correct statements from the following:

A: Densification of soil is a type of mechanical stabilisation.
B: Addition of calcium chloride can facilitate soil compaction.
C: Heating of soil is not a method of soil stabilisation.

1. B and C only
2. C only
3. A and B only
4. A only

Question ID: 4410097020 Opt 1: 44100927825 Opt 2: 44100927823 Opt 3: 44100927824 Opt 4: 44100927826

Q.41 Which of the following measures CANNOT prevent cracks resulting from sulphate attack?

1. Use of rich cement mortar or concrete in construction
2. Use of bricks containing high percentage of soluble salts
3. Use of sulphate resistant cement in construction
4. Provision of damp proofing to maintain dry conditions

Question ID: 44100932045 Opt 1: 441009127681 Opt 2: 441009127680 Opt 3: 441009127682 Opt 4: 441009127683

Q.42 Consider the following statements and choose the correct option:

Statement 1: Prolonged exposure to noise levels exceeding 85 dB can cause permanent hearing loss.
Statement 2: Temporary Threshold Shift (TTS) is permanent damage caused by short-term exposure to loud noise.

1. Statement 1 is true, but Statement 2 is false.
2. Both statements are true, and Statement 2 correctly explains Statement 1.
3. Both statements are true, but Statement 2 does not explain Statement 1.
4. Statement 1 is false, but Statement 2 is true.

`Question ID: 441009265036` `Opt 1: 4410091030186` `Opt 2: 4410091030184` `Opt 3: 4410091030185` `Opt 4: 4410091030187`

Q.43 In a penetration macadam type bituminous surfacing, what is the correct sequence of construction?

1. Placing fine aggregates before bitumen application
2. Premixing bitumen with aggregates, followed by compaction
3. Laying coarse aggregates, followed by bitumen application and spreading screenings
4. Applying bitumen first, then placing and compacting aggregates

`Question ID: 4410091114677` `Opt 1: 4410094394654` `Opt 2: 4410094394651` `Opt 3: 4410094394652` `Opt 4: 4410094394653`

Q.44 Which type of masonry junction is commonly used when future extension of a wall is anticipated?

1. Quoining junction
2. Cross-junction
3. Dado junction
4. Toothed junction

`Question ID: 4410091507243` `Opt 1: 4410095953517` `Opt 2: 4410095953518` `Opt 3: 4410095953519` `Opt 4: 4410095953520`

Q.45 A 5 kg sample of coarse aggregate is completely immersed in water for 24 hours. After surface drying, its weight is found to be 5.08 kg. If oven drying the sample reduces its weight back to 5 kg, what is the water absorption percentage of the aggregate?

1. 2.5%
2. 3%
3. 1.6%
4. 4%

Question ID: 4410099922 Opt 1: 44100939467 Opt 2: 44100939468 Opt 3: 44100939470 Opt 4: 44100939469

Q.46 In hydrological studies, interception is an important process because it:

1. Controls groundwater flow directly
2. Affects the amount of rainfall reaching the ground
3. Increases total rainfall
4. Has no effect on catchment hydrology

Question ID: 4410091371639 Opt 1: 4410095417954 Opt 2: 4410095417952 Opt 3: 4410095417953 Opt 4: 4410095417955

Q.47 What term describes a rectangular measurement system used to determine the position of points in the drawing area?

1. Absolute coordinates
2. Polar coordinates
3. Cartesian coordinates
4. Relative coordinates

Question ID: 4410091077310 Opt 1: 4410094246625 Opt 2: 4410094246626 Opt 3: 4410094246623 Opt 4: 4410094246624

Q.48 Read the two statements given below and select the most appropriate option:

Assertion (A): Coursed rubble masonry always gives the same finish quality as ashlar masonry.
Reason (R): In coursed rubble masonry, stones are roughly dressed and laid in leveled courses.

1. Both A and R are true, but R does not explain A
2. A is true, R is false
3. A is false, R is true
4. Both A and R are true, and R explains A

Question ID: 4410091507274 Opt 1: 4410095953650 Opt 2: 4410095953651 Opt 3: 4410095953652 Opt 4: 4410095953649

Q.49 Which of the following is the default rendering engine in Autodesk 3ds Max?

1. Scanline Renderer
2. V-Ray
3. Arnold
4. Mental Ray

Question ID: 44100968991 Opt 1: 441009274600 Opt 2: 441009274597 Opt 3: 441009274598 Opt 4: 441009274599

Q.50 What is the floor finish used in timber flooring called?

1. Boarding
2. Lapping
3. Tiling
4. Bridging

Question ID: 44100939715 Opt 1: 441009158389 Opt 2: 441009158392 Opt 3: 441009158390 Opt 4: 441009158391

Q.51 Which of the following statements is TRUE regarding non-structural cracks?

1. They pose a threat to the structural durability of a building.
2. Extensive cracks occur in foundations, slabs, etc.
3. They are caused by atmospheric changes on building materials.
4. Repairing and rectifying them is generally very difficult.

Question ID: 44100932453 Opt 1: 441009129281 Opt 2: 441009129283 Opt 3: 441009129284 Opt 4: 441009129282

Q.52 As per Indian Standard: 2470, a septic tank should have a minimum liquid capacity of:

1. 100 litres
2. 1000 litres
3. 10 litres
4. 1 litre

Question ID: 44100939598 Opt 1: 441009157915 Opt 2: 441009157913 Opt 3: 441009157916 Opt 4: 441009157914

Q.53 Select the INCORRECT factor affecting the durability of concrete.

1. Quantity of reinforcement
2. Water cement ratio
3. Weather exposure conditions
4. Use of superplasticizers

Question ID: 4410098150 Opt 1: 44100932319 Opt 2: 44100932317 Opt 3: 44100932316 Opt 4: 44100932318

Q.54 The resultant of a non-concurrent coplanar force system can be replaced by:

1. A single force
2. A single moment
3. A couple only
4. A force and a couple

Question ID: 4410091469861 Opt 1: 4410095807382 Opt 2: 4410095807385 Opt 3: 4410095807383 Opt 4: 4410095807384

Q.55 A solid, circular shaft is subjected to a pure twisting moment (torque) T. Which primary type of strain is developed on an elemental volume located on the outer surface of the shaft?

1. Volumetric strain
2. Longitudinal strain
3. Shear strain
4. Compressive strain

Question ID: 4410091372513 Opt 1: 4410095421485 Opt 2: 4410095421483 Opt 3: 4410095421484 Opt 4: 4410095421486

Q.56 Which grade of bitumen is widely used as a replacement for old 80/100 penetration grade in spraying applications like surface dressing and pavement construction in very cold weather, and is also used for manufacturing bitumen emulsions and modified bitumen products?

1. VG-30 Bitumen
2. VG-20 Bitumen
3. VG-10 Bitumen
4. VG-40 Bitumen

Question ID: 4410091191662 Opt 1: 4410094700772 Opt 2: 4410094700771 Opt 3: 4410094700770 Opt 4: 4410094700773

Q.57 Which of the following is the correct pair related to the application and classification of bricks?

1. Second-class bricks – Paving of roads
2. Third-class bricks – External exposed walls
3. Fourth-class bricks – Decorative facing works
4. First-class bricks – Load-bearing walls

Question ID: 4410091182707 Opt 1: 4410094664886 Opt 2: 4410094664887 Opt 3: 4410094664888 Opt 4: 4410094664885

Q.58 Which type of canal alignment generally requires minimum cross drainage works?

1. Side slope canal
2. Ridge canal
3. Inundation canal
4. Contour canal

Question ID: 4410091289723 Opt 1: 4410095090812 Opt 2: 4410095090810 Opt 3: 4410095090813 Opt 4: 4410095090811

Q.59 Which plane table method is used when the instrument is set up at a point whose location is not previously known on the map?

1. Intersection
2. Radiation
3. Traversing
4. Resection

Question ID: 441009269494 Opt 1: 4410091047171 Opt 2: 4410091047170 Opt 3: 4410091047173 Opt 4: 4410091047172

Q.60 For repair and waterproofing purposes, _____ is used to fill cracks, gaps, and cavities.

1. Tamping
2. Ramming
3. Curing
4. Grouting

Question ID: 44100932495 Opt 1: 441009129327 Opt 2: 441009129328 Opt 3: 441009129326 Opt 4: 441009129325

Q.61 Read the following statements and select the correct option:

Statement 1: A rigid body is an ideal material that undergoes no deformation under applied loads.
Statement 2: In reality, all materials behave as rigid bodies under all loading conditions.

1. Only Statement 2 is correct.
2. Both statements are correct.
3. Both statements are incorrect.
4. Only Statement 1 is correct.

Question ID: 441009108871 Opt 1: 441009433304 Opt 2: 441009433301 Opt 3: 441009433302 Opt 4: 441009433303

Q.62 The coefficient of permeability for a soil is 0.80 mm/s at a void ratio of 0.5. Find the permeability of the same soil at the same temperature, but with a void ratio of 0.25.

1. 0.2 mm/s
2. 1.6 mm/s
3. 0.12 mm/s
4. 0.4 mm/s

Question ID: 441009274470 Opt 1: 4410091066175 Opt 2: 4410091066177 Opt 3: 4410091066174 Opt 4: 4410091066176

Q.63 What does the silt carried by canals mainly consist of?

1. Fine soil particles carried by water
2. Chemical impurities
3. Organic matter in water
4. Large boulders

Question ID: 4410091216813 Opt 1: 4410094800661 Opt 2: 4410094800663 Opt 3: 4410094800662 Opt 4: 4410094800660

Q.64 Two thin rods of equal length and mass, one along the x-axis (horizontal) and the other along the y-axis (vertical), cross each other at mid-length. The moment of inertia about the z-axis perpendicular to the plane will be:

1. greater than the rod lying along the y-axis
2. equal for both rods combined
3. zero
4. greater than the rod lying along the x-axis

Question ID: 4410091208850 Opt 1: 4410094768895 Opt 2: 4410094768896 Opt 3: 4410094768897 Opt 4: 4410094768894

Q.65 Which of the following properties of bitumen is most critical in determining its performance at high pavement temperatures?

1. Penetration
2. Softening point
3. Flash point
4. Ductility

Question ID: 4410091600968 Opt 1: 4410096323551 Opt 2: 4410096323554 Opt 3: 4410096323552 Opt 4: 4410096323553

Q.66 In IS 456:2000, the limiting moment of resistance of a rectangular singly reinforced section is based on:

1. percentage of binding wires used
2. ultimate tensile strain in concrete
3. compression steel yield strength
4. limiting depth of neutral axis

Question ID: 4410091375763 Opt 1: 4410095434490 Opt 2: 4410095434493 Opt 3: 4410095434491 Opt 4: 4410095434492

Q.67 According to IS 456:2000, if the 28-day concrete cube strength falls below the required value, what is the next step?

1. Core test or rebound hammer test is conducted
2. The structure is rejected immediately
3. Additional curing is provided
4. The structure is demolished

Question ID: 441009128263 Opt 1: 441009509672 Opt 2: 441009509670 Opt 3: 441009509671 Opt 4: 441009509673

Q.68 Which of the following information is NOT provided by a building section drawing?

1. Foundation used for the structure
2. Height of the structure
3. Length and breadth of rooms
4. Type of material used for the structure

Question ID: 44100936227 Opt 1: 441009144435 Opt 2: 441009144434 Opt 3: 441009144436 Opt 4: 441009144433

Q.69 Which of the following is a main combustible component of biogas?

1. Carbon dioxide
2. Butane
3. Hydrogen sulfide
4. Methane

Question ID: 4410096546 Opt 1: 44100925938 Opt 2: 44100925937 Opt 3: 44100925936 Opt 4: 44100925935

Q.70 In a Queen Post Truss Roof, which of the following statements is FALSE?

1. It has a straining beam and straining sill.
2. It has two tension members.
3. It is suitable for spans of 8 to 12 m.
4. It has one central vertical post.

Question ID: 44100939729 Opt 1: 441009158448 Opt 2: 441009158446 Opt 3: 441009158445 Opt 4: 441009158447

Q.71 Epoxy injection method is effectively used for repairing which type of cracks?

1. Dormant cracks
2. Active cracks
3. Moving cracks
4. Expanding cracks

Question ID: 44100932528 Opt 1: 441009129365 Opt 2: 441009129366 Opt 3: 441009129367 Opt 4: 441009129368

Q.72 Identify the INCORRECT statement about the principles of triangulation:

1. The stations at which astronomical observations are made are called geodetic stations.
2. The line whose length is known is called base line.
3. To control the accumulation of errors in triangulation, subsidiary bases are selected.
4. The system of interconnected triangles is called the triangulation system.

Question ID: 4410091399101 Opt 1: 4410095526123 Opt 2: 4410095526121 Opt 3: 4410095526124 Opt 4: 4410095526122

Q.73 In railway track construction, which method is most commonly used for laying new tracks?

1. Telescopic Method
2. Trenchless Method
3. Cut-and-Cover Method
4. Incremental Launching

Question ID: 44100959240 Opt 1: 441009235861 Opt 2: 441009235862 Opt 3: 441009235864 Opt 4: 441009235863

Q.74 In the method of sections as applicable to analysis of plane trusses, the number of unknown member forces that can be determined in one section will not exceed:

1. 3
2. 4
3. 6
4. 2

Question ID: 4410091273265 Opt 1: 4410095025077 Opt 2: 4410095025078 Opt 3: 4410095025079 Opt 4: 4410095025076

Q.75 What contributes to the pollution of surface water bodies?

1. Natural springs
2. Community waste
3. Rainwater harvesting
4. Underground aquifers

Question ID: 441009942891 Opt 1: 4410093725214 Opt 2: 4410093725215 Opt 3: 4410093725217 Opt 4: 4410093725216

Q.76 The moment distribution method used in structural analysis is best described as:

1. An iterative stiffness based method for indeterminate beams and frames
2. An energy-based method for determinate beams and frames
3. A matrix-based method for simply supported beam systems
4. A force-based method for pin-jointed truss systems

Question ID: 4410091420792 Opt 1: 4410095612092 Opt 2: 4410095612093 Opt 3: 4410095612095 Opt 4: 4410095612094

Q.77 For a singly reinforced RCC beam under flexure, when A_{st} provided > A_{st} required for a balanced section, the beam is expected to fail by:

1. Torsional instability
2. Shear cracking followed by collapse
3. Brittle crushing of concrete in compression
4. Ductile yielding of steel in tension

Question ID: 4410091291163 Opt 1: 4410095096537 Opt 2: 4410095096535 Opt 3: 4410095096536 Opt 4: 4410095096534

Q.78 A rectangular lamina has a width of 200 mm and height of 300 mm. Using standard formulas, find its moment of inertia in mm^4 about the centroidal x-axis (parallel to width).

1. 45×10^8 mm^4
2. 4.5×10^6 mm^4
3. 4.5×10^8 mm^4
4. 5×10^8 mm^4

Question ID: 44100979552 Opt 1: 441009316014 Opt 2: 441009316012 Opt 3: 441009316011 Opt 4: 441009316013

Q.79 Which of the following sections is most suitable for long compression members to prevent local buckling?

1. Solid rectangular section
2. Flat plate section
3. Solid circular section
4. Built-up I-section with batten or lacing

Question ID: 4410091339112 Opt 1: 4410095288531 Opt 2: 4410095288532 Opt 3: 4410095288530 Opt 4: 4410095288529

Q.80 In a Free Body Diagram (FBD), how is tension in a string represented?

1. By an arrow pointing towards the object along the direction of the string
2. By an arrow pointing perpendicular to the string
3. By an arrow pointing away from the object along the direction of the string
4. By an arrow pointing downward

Question ID: 44100962691 Opt 1: 441009249825 Opt 2: 441009249827 Opt 3: 441009249824 Opt 4: 441009249826

Q.81 Which type of plastering is best suited for providing thermal insulation in rural houses?

1. Gypsum plastering
2. Mud plastering
3. Cement plastering
4. Stucco plastering

Question ID: 4410091583733 Opt 1: 4410096256257 Opt 2: 4410096256259 Opt 3: 4410096256256 Opt 4: 4410096256258

Q.82 Match the following types of dump trucks with their descriptions:

Dump Truck	Description
1. Truck trailer	A. A truck trailer with its entire dead weight self-supported
2. Semi-trailer	B. A truck trailer designed to reduce load on tractor, having ≥ 1 axle, fifth wheel, drawbar
3. Full trailer	C. A truck trailer with ≥ 1 axle whose front part rests on tractor
4. Load dividing dolly	D. A vehicle (with or without auxiliary power) used to redistribute weight

1. 1-A, 2-B, 3-C, 4-D
2. 1-C, 2-D, 3-A, 4-B
3. 1-D, 2-D, 3-A, 4-B
4. 1-B, 2-C, 3-D, 4-A

Question ID: 4410091201405 Opt 1: 4410094739404 Opt 2: 4410094739406 Opt 3: 4410094739407 Opt 4: 4410094739405

Q.83 A group of piles are connected by an RCC element known as:

1. Strap beam
2. Raft
3. Pile cap
4. Plinth beam

`Question ID: 44100945155` `Opt 1: 441009179831` `Opt 2: 441009179832` `Opt 3: 441009179829` `Opt 4: 441009179830`

Q.84 In pump analysis, the sum of suction head and delivery head is called:

1. Manometric head
2. Pressure head
3. Frictional head
4. Static head

`Question ID: 4410091044864` `Opt 1: 4410094128565` `Opt 2: 4410094128564` `Opt 3: 4410094128562` `Opt 4: 4410094128563`

Q.85 What is a potential drawback of using High Performance Concrete in structural applications?

1. Increased susceptibility to chemical reactions in marine environment
2. Low bond strength with reinforcement
3. High consistency makes achieving proper compaction difficult
4. Lower strength compared to conventional concrete

`Question ID: 44100924829` `Opt 1: 44100999040` `Opt 2: 44100999039` `Opt 3: 44100999037` `Opt 4: 44100999038`

Q.86 Identify the INCORRECT statement about standard measurement books:

1. All standard measurement books are reviewed in a year.
2. It is helpful in preparing repair estimates.
3. It is preserved for 10 years only.
4. It is numbered alphabetically.

`Question ID: 44100922716` `Opt 1: 44100990594` `Opt 2: 44100990591` `Opt 3: 44100990593` `Opt 4: 44100990592`

Q.87 For cohesive clayey soils in subgrade, the most effective method of stabilization is:

1. Cement stabilization
2. Mechanical stabilization
3. Lime stabilization
4. Bituminous stabilization

Question ID: 4410091600881 Opt 1: 4410096323204 Opt 2: 4410096323203 Opt 3: 4410096323206 Opt 4: 4410096323205

Q.88 Which of the following kerb marking schemes is INCORRECTLY matched?

1. Black and White – Visibility enhancement
2. Yellow and Black – Parking restrictions
3. Green and White – Indicates national highways
4. Red and White – Special no-parking zones

Question ID: 4410091251868 Opt 1: 4410094940586 Opt 2: 4410094940587 Opt 3: 4410094940589 Opt 4: 4410094940588

Q.89 If orientation of the plane table is lost during the survey, which method is most suitable to reorient it?

1. By estimation
2. Back sighting method
3. By re-plotting the entire survey
4. By ignoring the orientation error

Question ID: 4410091251923 Opt 1: 4410094940811 Opt 2: 4410094940810 Opt 3: 4410094940812 Opt 4: 4410094940813

Q.90 In a vertical single column manometer, the orientation of the column is kept:

1. inclined at a certain angle
2. horizontal
3. exactly vertical to improve sensitivity
4. bent in a loop

Question ID: 4410091213790 Opt 1: 4410094788566 Opt 2: 4410094788568 Opt 3: 4410094788567 Opt 4: 4410094788569

Q.91 Find the cost of plastic strainers required in a tube well of 30 mm diameter and 30 m depth fitted with a hydraulic hand pump. A medium slot size strainer of 200 cm length is installed in the water-bearing aquifer layer at 20 m below ground level, and another high slot size strainer of 300 cm length is installed in the water-bearing layer at 30 m depth. Assume the cost rate for both strainer slot sizes is ₹350/meter.

1. ₹1,600/-
2. ₹1,750/-
3. ₹1,900/-
4. ₹600/-

Question ID: 4410091513741 Opt 1: 4410095978933 Opt 2: 4410095978931 Opt 3: 4410095978932 Opt 4: 4410095978930

Q.92 In an RCC beam, what type of reinforcement is provided to resist shear?

1. Compression bars
2. Longitudinal bars
3. Lateral ties
4. Stirrups

Question ID: 44100918918 Opt 1: 44100975446 Opt 2: 44100975444 Opt 3: 44100975447 Opt 4: 44100975445

Q.93 'Reinforced concrete walls cantilevering vertically from the base designed to resist lateral forces' applies to which of the following terms?

1. Shear wall
2. Panel wall
3. Load bearing wall
4. Partition wall

Question ID: 441009307414 Opt 1: 4410091196328 Opt 2: 4410091196331 Opt 3: 4410091196329 Opt 4: 4410091196330

Q.94 Which of the following is a key difference between short columns and long columns?

1. Short columns fail due to buckling, while long columns fail due to crushing.
2. Long columns experience pure axial compression, while short columns experience lateral deflection.
3. Long columns have a higher slenderness ratio compared to short columns.
4. Short columns are only used in steel structures, while long columns are used in concrete structures.

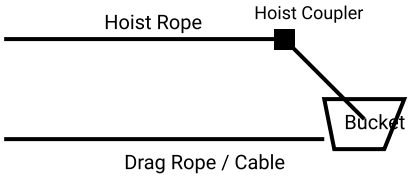
Question ID: 44100912228 Opt 1: 44100948715 Opt 2: 44100948718 Opt 3: 44100948716 Opt 4: 44100948717

Q.95 Which of the following pairs is CORRECTLY matched regarding door fittings?

1. Hinges – Used for locking doors
2. Door Stopper – Used to hold door at an angle
3. Aldrop – Used for sliding doors
4. Tower Bolt – Used for automatic door opening

Question ID: 44100966817 Opt 1: 441009266066 Opt 2: 441009266065 Opt 3: 441009266068 Opt 4: 441009266067

Q.96 Identify the earthwork equipment shown in the schematic diagram below:



1. Forward Loader
2. Grab
3. Dragline
4. Conveyor

Question ID: 44100981385 Opt 1: 441009323097 Opt 2: 441009323098 Opt 3: 441009323099 Opt 4: 441009323096

Q.97 In CAD, which command is used to remove a portion of an object that extends beyond a boundary?

1. Rotate
2. Extend
3. Trim
4. Copy

Question ID: 4410091319211 Opt 1: 4410095208407 Opt 2: 4410095208404 Opt 3: 4410095208405 Opt 4: 4410095208406

Q.98 What is an agreement enforced by law between two parties called?

1. Settlement
2. Tender
3. Contract
4. Contractor

Question ID: 441009997367 Opt 1: 4410093941448 Opt 2: 4410093941445 Opt 3: 4410093941446 Opt 4: 4410093941447

Q.99 Which of the following is a natural method of timber seasoning where wood is dried by exposing it to air?

1. Kiln seasoning
2. Water seasoning
3. Chemical seasoning
4. Air seasoning

Question ID: 4410091086080 Opt 1: 4410094281867 Opt 2: 4410094281868 Opt 3: 4410094281870 Opt 4: 4410094281869

Q.100 Find the hydraulic radius (m) for a rectangular channel section having width (b) and depth (d).

1. m = (b + 2d) / (b × d)
2. m = d / (b + 2d)
3. m = (b × d) / (b + 2d)
4. m = b / (b + 2d)

Question ID: 4410091519287 Opt 1: 4410096000406 Opt 2: 4410096000408 Opt 3: 4410096000409 Opt 4: 4410096000407